

Вопросы эпиграфики

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Выпуск VII
Часть 1

Материалы I Международной конференции «Вопросы эпиграфики»

Университет
Дмитрия Пожарского

2013

Печатается по решению Учёного совета
Университета Дмитрия Пожарского

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Вопросы эпиграфики. Выпуск 7. Часть 1. [Сб. ст.] / Университет Дмитрия Пожарского / Отв. ред. А. Г. Авдеев. — М.: Русский Фонд Содействия Образованию и Науке. 2013. — 688 с.

ISBN 978-5-91244-116-5 (часть 1)

Первый выпуск сборника, составленного на основе материалов I Международной конференции «Вопросы эпиграфики», состоит из трёх разделов. Первый раздел посвящён эпиграфике Востока. Во втором разделе публикуются статьи по античной эпиграфике. Третий раздел посвящён эпиграфике доколумбовой Америки. В приложении даны краткие сведения об авторах и список сокращений.

ISBN 978-5-91244-116-5 (часть 1)

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Positional Analysis of Indus Signs¹

A. Fuls

The Indus culture (2500 to 1800 BC) developed a writing system which remains undeciphered until today. The inscriptions found on different artefacts like seals, tablets, pots, bangles, tags, and other types show a distinct usage of Indus Signs. Some signs are typically found in initial position, while other are mostly terminal, but many signs were only be classified as somehow medial, which is not a precise term. A new method is developed to classify each signs positional behaviour and to differentiate between different medial positions. This allows a more detailed analysis and classification of Indus signs for all texts as well as for specific text classes. The method can also be used to show that often graphically similar signs differ in their preferred positional behaviour and are not allographs but distinct signs. The positional histograms of signs derived from different text classes indicate, that Indus signs can have more than one preferred text position depending on the content and/or the structure of the texts. This is important to emphasis, because very often previous statistical tests of Indus signs have put all texts together without considering the possibility of dealing with different context or text structures.

1. Introduction

Since the excavations in Harappa and Mohenjo-daro Indus signs are tabulated and distinguished by graphic features (graphems). This results into sign lists published by I. Mahadevan² (419 signs) and A. Parpola³ (386 signs), among others. The question remains which signs are distinct signs and which signs are allographs with the same meaning. The identification of allographs is a time consuming task by searching for sign replacement pattern and requires a large text corpus. This can be

¹ Many thanks to Bryan K. Wells and Chandrasekhar Subramanian for the corrections and suggestions to an earlier version of the paper.

² *Mahadevan I.* The Indus Script, Texts Concordance and Tables // Archaeological Survey of India. New Delhi, 1977.

³ *Parpola A.* Deciphering the Indus Script. Cambr., 1994.

done by the structure analysis (developed by E. O. Forrer⁴) and was applied by B. Wells⁵ to create a detailed sign list with 676 signs. This looks like a retrograde development as one is looking for a condensed sign list, but as long as allographs cannot be distinguished from distinct signs this is the most secure way, especially for low frequency signs. It is the future goal to reduce the sign list step by step analyzing the sign's behaviour individually to identify allographs and graphic variants.

The following research is based on an updated sign list from B. Wells (2006) with 695 signs (last update autumn 2009). Inscription are stored in the ICIT database (Interactive Corpus of Indus Texts) with a total of 3898 inscribed artefacts (one to three sided) and 4791 texts. 3723 texts of them are complete with 13556 signs. They are used as the text corpus for the following analysis of sign positions.

Many artefacts have identical texts coming from the same site and often they come from the same mold. Other duplicate texts have the same provenience. These artefacts are often incised or cooper miniature tablets. Because they are often found on tablets, the effect of duplicate texts is called the TAB effect⁶. In the following analysis identical texts of the same artefact type coming from the same site are reduced to one text. Otherwise they would lead to a high frequency of some signs, which do not represent the usage of signs in the Indus script but the circumstances influenced by excavations. Additionally, multiple line texts are not included, since it is uncertain how to read the second or third line. The lines could be read all right/left, boustrophedon, or each line represents an independent phrase⁷. This leaves 2474 complete and different texts, reducing also the number of relevant signs to 9700 signs altogether.

The mean text length are 4.01 signs. Table 1 summarizes the text lengths of the study data. They are stored in an online database of Indus texts⁸.

⁴ Forrer E. O. Die hethitische Bilderschrift. Chicago, 1932.

⁵ Wells B. Epigraphic Approaches to Indus Writing. Dissertation. Harvard University, 2006. P. 67–93.

⁶ Idem. P. 168.

⁷ Parpola A. Deciphering... P. 64–67.

⁸ Fuls A. Entwicklung einer geographisch-epigraphischen Datenbank der Indusschrift // Entwicklerforum Geodäsie und Geoinformationstechnik / Sven Weisbrich and Robert Kaden (ed.). Aachen, 2010. P. 29–45.

To get a better idea about the Indus writing system and the fraction of logograms versus syllables I have made a comparison of sign frequency histograms between different writing systems (logographic, syllabic, logosyllabic, alphabet with/without vowels). The result is that the Indus writing system is most similar to Proto-Sumerian cuneiform, an early stage with many logograms and about 37% to 54% syllables. A regression line between their sign frequencies shows a strong relationship of $R^2 = 0.9986$ (Wells 2006, p. 96). Therefore we can expect about the same fraction of syllables in the Indus script.

Table 1

**Frequency of text lengths in the reduced text corpus
of complete inscriptions without duplicates
from the same site or artefact type**

text length	total № of texts	№ of texts from Mohenjo-daro	№ of texts from Harappa
1	232	80	112
2	477	163	244
3	477	195	225
4	412	188	177
5	360	212	110
6	233	148	57
7	150	107	32
8	59	41	9
9	40	24	10
10	19	11	5
11	7	6	1
12	4	3	1
13	3	2	1

2. Methods of Sign Analysis

Previous works has already distinguished between initial, medial, and terminal sign position. It is well known that some signs like sign 740 are mostly located in the terminal position of texts, while others are very often in initial position of texts. The preferred sign positions gives

us some clues about the possible sign function assuming a language with a fixed syntax. We can expect that logograms describing objects, names, or actions have a limited text position, while syllables can appear more diverse text positions. Some signs might be polyvalent. The distribution of a sign position can be used to evaluate possible sign functions.

B. Wells⁹ classifies some signs according to their preferred text position as Initial signs (INS), Terminal Marker (TMK), and Post-Terminal Marker (PTM). He also discovered, that sign 60 appears only in position 1 to 4 (except 5 two-lined texts), which is often replaced by sign 1 or sign 2¹⁰. They are called Initial Cluster Terminal Marker (ICTM). N. Yadav et al.¹¹ analyze combinations of two to four signs in the Indus script and their frequency in solo, left, middle, and right position. Conclusions derived from high frequencies in solo, left-end or right-end position are well defined, but sign combinations in middle position can not be interpreted.

One simple approach is to label signs as being initial, medial or terminal. This results in a loss of all details concerning medial signs. In order to improve our understanding of the behaviour of medial signs a new method is developed, which gives us a more detailed impression on medial sign positions.

The question is, how to classify signs that are commonly medial and how to distinguish different sign positions in the middle part of texts. Because of the different text lengths medial/middle is not a precise term. This is a reason to develop a normalized weighted sign position.

2.1 Normalised Weighted Sign Position

To compare sign positions of different text length it is necessary to normalise the text length. Normalisation is performed by reducing long texts and stretching short texts to a standard length. Here, as a standard norm a text length of 10 signs is used. The normalisation is linear as shown in Figure 1.

⁹ Wells B. *Epigraphic Approaches...* P. 138.

¹⁰ *Idem.* P. 141.

¹¹ Yadav N., Vahai M. N., Mahadevan I., Joglekar H. A statistical approach for pattern search in Indus writing // *International Journal of Dravidian Linguistics*. 2008. Vol. 37 (1). P. 9–13.

The relative sign position can be calculated as follows. The original sign position SP of a text with L signs is scaled to a normalised text length NL (e.g. NL=10).

$$\text{MINP} = \text{int}((\text{SP} - 1) \times \text{NL}/\text{L} + 1) \quad (1)$$

$$\text{MAXP} = \text{int}(\text{SP} \times \text{NL}/\text{L}) \quad (2)$$

with

SP: absolute sign position

L: text length

NL: normalised text length (here NL = 10 signs)

MINP: minimum of relative text position

MAXP: maximum of relative text position

The results MINP and MAXP are rounded down to integer values in the range [1,NL]. They represent the minimum and maximum value of the relative sign position(s).

After normalisation a sign position of a short text might be counted more than once (like sign 002 falling into relative sign position 9 and 10, Fig. 1). To avoid an overweighting of signs from short texts each sign position is weighted linearly to text length. This means, that signs of short texts get a lower weight than signs of longer texts.

The weighting is calculated as follows:

$$W = \text{L}/\text{NL} \quad (3)$$

L: text length

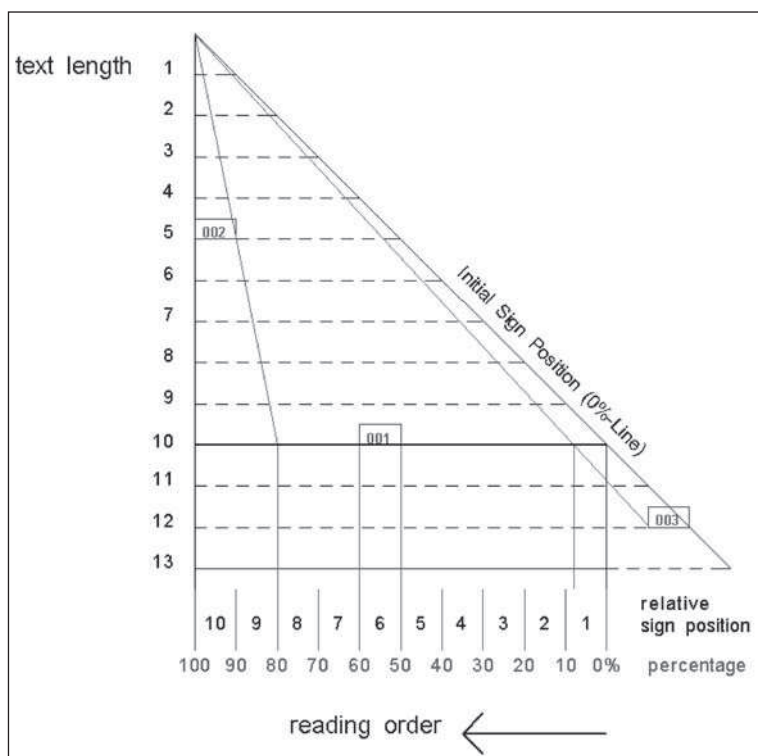
NL: normalised text length

W: weight

This means, that each sign in a text of NL signs gets a weight of 1, while longer texts have a higher precision of each sign position and get a higher weight. In contrast, shorter texts get a smaller weight, since their sign position is less precise.

Fig. 1

Normalisation of sign positions for different text lengths. Sign 001 falls between 50-60% (relative position 6), sign 002 falls between 80-100% (relative position 9+10), and sign 003 falls between 0-8.3% (relative position 1). Thereby, the range of each relative sign position (MINP und MAXP) depends on the text length and is normalised to a text length of 10 signs.



2.2 Histogram

The histogram shows the frequency of sign positions for a constant number of intervals. Each interval represents a normalised sign position counting from 1 to NL. In the following discussion NL is set to 10. To calculate the histogram for each sign the weight is added to the

relative sign position(s) between MINP and MAXP. For example, sign 002 from a text of length 5 has a weight of 0.5 and is counted twice, for relative sign position 9 and 10, respectively. In contrast, sign 003 from a text of length 12 has a weight of 1.2, which is added to position 1 (Fig. 1).

Some signs are very frequent producing high values in the histogram. For visual reasons the height of the histogram is scaled to the same size, so that the histogram of rare and frequent signs can be easily compared.

To analyze sign positions only complete texts can be used. Therefore incomplete texts are omitted. Additionally, there are many duplicates of identical texts often found at the same site or even at the same mold. Reducing them to one text results into a list of 2474 complete and different texts.

3. Classification of Indus Signs

Many Indus signs are already recognised as initial or terminal signs¹². It can be shown that histograms of normalised weighted sign positions are useful to distinguish between common terminal and initial signs. Initial and terminal signs can easily be compared against graphically similar signs to check if their preferred positions are identical or not.

3.1 Initial signs

Initial signs are known from structural analysis and appear very often in first position¹³. Most but not all of their histograms conform to this pattern (Table 2, left column).

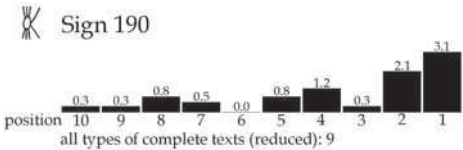
Signs 690, 790, 824, 850, and 921 are graphically similar to the Initial signs 692, 817, 820, 861, and 921, respectively. But the positional frequency shows, that they locate in different positions, not exclusively initial. This confirms, that they are distinct signs despite their graphic similarity (Table 2, right column). Other signs mostly initial are signs 190, 853, 880 (Fig. 2).

¹² Mahadevan I. *The Indus Script...*; Parpola A. *Deciphering...*; Wells B. *Epigraphic Approaches...*

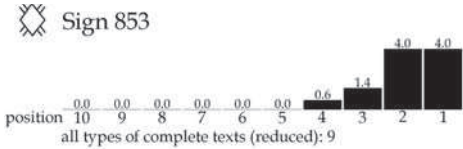
¹³ Wells B. *Epigraphic Approaches...* P. 138.

Fig. 2

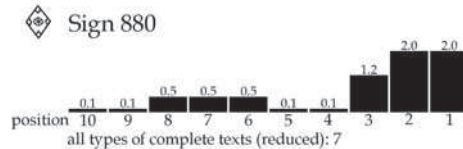
Initial signs 190, 853, and 880



(a) Sign 190



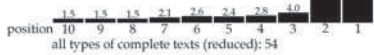
(b) Sign 853



(c) Sign 880

Table 2

Positional histograms of Initial sign and graphically similar signs

Initial Sign	Similar sign but not exclusively initial																																												
 Sign 692 <table><tr><th>position</th><th>10</th><th>9</th><th>8</th><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th></tr><tr><td>all types of complete texts (reduced): 54</td><td>1.5</td><td>1.5</td><td>1.5</td><td>2.1</td><td>2.6</td><td>2.4</td><td>2.8</td><td>3.0</td><td>10.9</td><td>10.9</td></tr></table>	position	10	9	8	7	6	5	4	3	2	1	all types of complete texts (reduced): 54	1.5	1.5	1.5	2.1	2.6	2.4	2.8	3.0	10.9	10.9	 Sign 690 <table><tr><th>position</th><th>10</th><th>9</th><th>8</th><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th></tr><tr><td>all types of complete texts (reduced): 90</td><td>6.4</td><td>11.6</td><td>26.6</td><td>11.4</td><td>11.4</td><td>6.1</td><td>8.7</td><td>4.0</td><td>7.8</td><td>8.2</td></tr></table>	position	10	9	8	7	6	5	4	3	2	1	all types of complete texts (reduced): 90	6.4	11.6	26.6	11.4	11.4	6.1	8.7	4.0	7.8	8.2
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Initial Sign	Similar sign but not exclusively initial																																												
◊ Sign 817 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 169 <table border="1"> <caption>Data for Sign 817</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>2.1</td></tr> <tr><td>9</td><td>2.2</td></tr> <tr><td>8</td><td>1.9</td></tr> <tr><td>7</td><td>2.8</td></tr> <tr><td>6</td><td>1.8</td></tr> <tr><td>5</td><td>5.8</td></tr> <tr><td>4</td><td>10.6</td></tr> <tr><td>3</td><td>75.5</td></tr> <tr><td>2</td><td>68.3</td></tr> <tr><td>1</td><td>69.3</td></tr> </tbody> </table>	Position	Frequency	10	2.1	9	2.2	8	1.9	7	2.8	6	1.8	5	5.8	4	10.6	3	75.5	2	68.3	1	69.3	◊ Sign 790 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 42 <table border="1"> <caption>Data for Sign 790</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>4.8</td></tr> <tr><td>9</td><td>2.1</td></tr> <tr><td>8</td><td>5.7</td></tr> <tr><td>7</td><td>6.6</td></tr> <tr><td>6</td><td>4.1</td></tr> <tr><td>5</td><td>4.1</td></tr> <tr><td>4</td><td>4.0</td></tr> <tr><td>3</td><td>4.0</td></tr> <tr><td>2</td><td>4.1</td></tr> <tr><td>1</td><td>4.1</td></tr> </tbody> </table>	Position	Frequency	10	4.8	9	2.1	8	5.7	7	6.6	6	4.1	5	4.1	4	4.0	3	4.0	2	4.1	1	4.1
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◊ Sign 820 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 159 <table border="1"> <caption>Data for Sign 820</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>5.6</td></tr> <tr><td>9</td><td>10.2</td></tr> <tr><td>8</td><td>7.2</td></tr> <tr><td>7</td><td>3.6</td></tr> <tr><td>6</td><td>4.6</td></tr> <tr><td>5</td><td>7.8</td></tr> <tr><td>4</td><td>9.4</td></tr> <tr><td>3</td><td>17.0</td></tr> <tr><td>2</td><td>57.8</td></tr> <tr><td>1</td><td>90.9</td></tr> </tbody> </table>	Position	Frequency	10	5.6	9	10.2	8	7.2	7	3.6	6	4.6	5	7.8	4	9.4	3	17.0	2	57.8	1	90.9	◊ Sign 824 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 23 <table border="1"> <caption>Data for Sign 824</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>4.6</td></tr> <tr><td>9</td><td>4.6</td></tr> <tr><td>8</td><td>3.1</td></tr> <tr><td>7</td><td>0.7</td></tr> <tr><td>6</td><td>0.7</td></tr> <tr><td>5</td><td>0.1</td></tr> <tr><td>4</td><td>1.0</td></tr> <tr><td>3</td><td>1.4</td></tr> <tr><td>2</td><td>3.1</td></tr> <tr><td>1</td><td>3.1</td></tr> </tbody> </table>	Position	Frequency	10	4.6	9	4.6	8	3.1	7	0.7	6	0.7	5	0.1	4	1.0	3	1.4	2	3.1	1	3.1
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◊ Sign 861 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 204 <table border="1"> <caption>Data for Sign 861</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>8.7</td></tr> <tr><td>9</td><td>9.4</td></tr> <tr><td>8</td><td>9.8</td></tr> <tr><td>7</td><td>12.2</td></tr> <tr><td>6</td><td>15.0</td></tr> <tr><td>5</td><td>7.4</td></tr> <tr><td>4</td><td>11.2</td></tr> <tr><td>3</td><td>24.7</td></tr> <tr><td>2</td><td>63.1</td></tr> <tr><td>1</td><td>62.1</td></tr> </tbody> </table>	Position	Frequency	10	8.7	9	9.4	8	9.8	7	12.2	6	15.0	5	7.4	4	11.2	3	24.7	2	63.1	1	62.1	◊ Sign 850 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 34 <table border="1"> <caption>Data for Sign 850</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>1.9</td></tr> <tr><td>9</td><td>1.6</td></tr> <tr><td>8</td><td>8.9</td></tr> <tr><td>7</td><td>9.2</td></tr> <tr><td>6</td><td>2.4</td></tr> <tr><td>5</td><td>3.1</td></tr> <tr><td>4</td><td>2.1</td></tr> <tr><td>3</td><td>1.5</td></tr> <tr><td>2</td><td>3.1</td></tr> <tr><td>1</td><td>3.1</td></tr> </tbody> </table>	Position	Frequency	10	1.9	9	1.6	8	8.9	7	9.2	6	2.4	5	3.1	4	2.1	3	1.5	2	3.1	1	3.1
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3	24.7																																												
2	63.1																																												
1	62.1																																												
Position	Frequency																																												
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3	1.5																																												
2	3.1																																												
1	3.1																																												
◊ Sign 920 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 107 <table border="1"> <caption>Data for Sign 920</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>2.5</td></tr> <tr><td>9</td><td>3.9</td></tr> <tr><td>8</td><td>4.8</td></tr> <tr><td>7</td><td>6.6</td></tr> <tr><td>6</td><td>5.5</td></tr> <tr><td>5</td><td>16.7</td></tr> <tr><td>4</td><td>3.6</td></tr> <tr><td>3</td><td>11.3</td></tr> <tr><td>2</td><td>90.4</td></tr> <tr><td>1</td><td>92.5</td></tr> </tbody> </table>	Position	Frequency	10	2.5	9	3.9	8	4.8	7	6.6	6	5.5	5	16.7	4	3.6	3	11.3	2	90.4	1	92.5	◊ Sign 921 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 15 <table border="1"> <caption>Data for Sign 921</caption> <thead> <tr><th>Position</th><th>Frequency</th></tr> </thead> <tbody> <tr><td>10</td><td>1.8</td></tr> <tr><td>9</td><td>1.7</td></tr> <tr><td>8</td><td>0.7</td></tr> <tr><td>7</td><td>2.1</td></tr> <tr><td>6</td><td>2.8</td></tr> <tr><td>5</td><td>1.7</td></tr> <tr><td>4</td><td>1.1</td></tr> <tr><td>3</td><td>0.7</td></tr> <tr><td>2</td><td>0.7</td></tr> <tr><td>1</td><td>3.0</td></tr> </tbody> </table>	Position	Frequency	10	1.8	9	1.7	8	0.7	7	2.1	6	2.8	5	1.7	4	1.1	3	0.7	2	0.7	1	3.0
Position	Frequency																																												
10	2.5																																												
9	3.9																																												
8	4.8																																												
7	6.6																																												
6	5.5																																												
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4	1.1																																												
3	0.7																																												
2	0.7																																												
1	3.0																																												

3.2 Terminal Markers

Terminal markers are discussed by Wells (2006, p. 198–200). They appear often at the end of texts. About half of them show a high frequency at the left side (terminal position), but many Terminal marker have a very different distribution of positional frequency (Tab. 3). Sign 156 and 158 appear anywhere at the second half of text positions. Sign 741 is similar to sign 740 with a maximum at position 5 but not terminal as sign 740.

Table 3

Positional histograms of Terminal marker and graphically similar signs

Terminal Markers	Similar signs not exclusively terminal
Sign 740 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 1329	Sign 741 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 141
Sign 520 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 248	
Sign 527 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 52	Sign 526 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 12
Sign 151 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 66	Sign 158 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 27
Sign 154 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 35	Sign 156 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 96
Sign 161 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 11	
Sign 565 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 11	Sign 564 position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 5

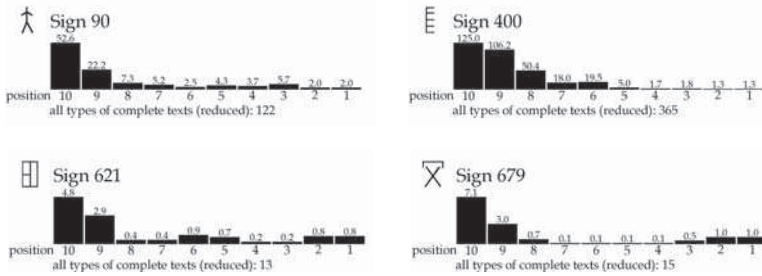
3.3 Post-Terminal Signs

Post-Terminal signs are discussed by B. Wells¹⁴. They appear mostly at the end of texts as an affix after Terminal markers. Post-Terminal signs are sign 90, 400, 621, and 679 (Fig. 3). Other Terminal or Post-Terminal signs are sign 161, 422, and 423, but their frequency is very low.

A high frequency focused on one position might be a good indicator of a grammatical function with a restricted grammatical position (e.g. case marker).

Fig. 3

Positional histograms of Post-Terminal signs

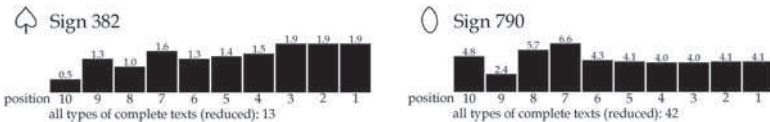


3.4 Constant sign distribution

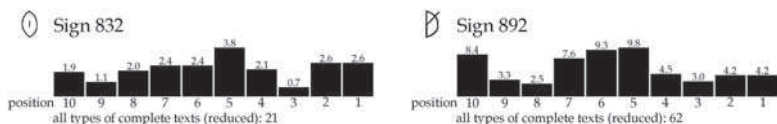
A constant frequency for all positions is a good indicator of a syllable, since syllables are used anywhere in the text without a syntactic preference. Therefore, signs with a constant distribution are good candidates of a syllable. Signs with an almost constant distribution are signs 382, 790, 832, and 892 (Fig. 4).

Fig. 4

Signs with an almost constant positional distribution



¹⁴ Wells B. Epigraphic Approaches... P. 198–200.

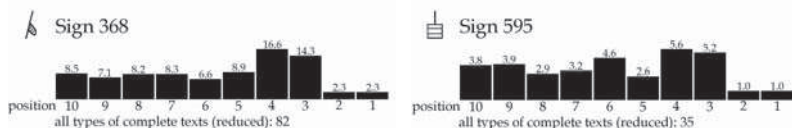


3.5 Constant but not initial sign distribution

Sign 368 and 595 are more or less constant but almost never initial (Fig. 5). The reason for the relative low frequency in initial position is unknown. It might indicate a syllabic value of a grammatical feature, which doesn't occur in initial position because the grammatical marker is always suffixed. Sign 368 is already recognized by Y. Knorozov¹⁵ as a grammatical marker.

Fig. 5

Signs with a constant but almost non initial sign distribution



3.7 Medial Sign Classification

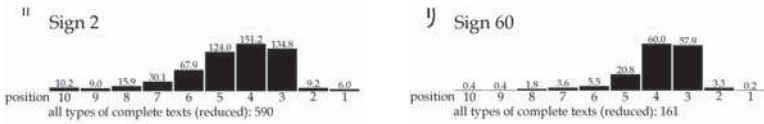
In the following sections three new classes of signs in preferred medial position are defined. They are called Early-Medial, Mid-Medial, and Late-Medial signs.

Signs which have a maximum at positions 3 to 4 are called Early-Medial signs (Fig. 6). Sign 2 and 60 are known as a marker after initial signs or initial sign clusters. As expected their preferred position is in the early medial text part. Wells (2006) calls them, together with sign 1, Initial Cluster Terminal Marker (ICTM), since they mark the end of initial clusters.

¹⁵ Knorozov Y. The formal analysis of the proto-indian texts. The Soviet Decipherment of the Indus Valley Script // A. Zide and K. Zvelebil (ed.). Janua linguarum. Series practica. № 156 The Hague, Mouton 1976. P. 97–107.

Fig. 6

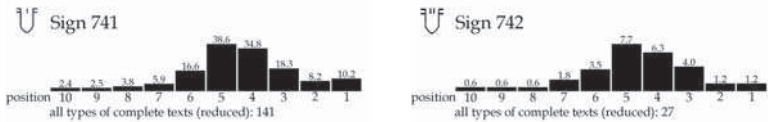
Signs with an early medial positional distribution



Signs which have a maximum at positions 5 to 6 are called Mid-Medial signs. The histogram of signs 741 and 742 show a maximum at position 5 (Fig. 7). Their positional behaviour differ from the typical Terminal sign 740, although they are graphically similar except the additional stroke(s). This shows, that markings can effect the preferred sign position.

Fig. 7

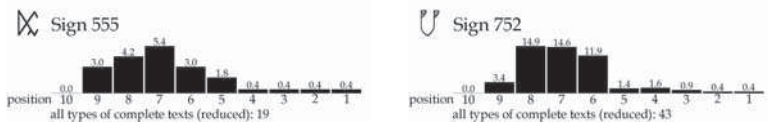
Signs 741 and 742 have a mid medial distribution



Signs which have a maximum at positions 7 to 8 are called Late-Medial signs, e.g. sign 555 and 752 (Fig. 8). Another Late-Medial sign is sign 590, but only in long patterned texts (see section 7, Fig. 9c).

Fig. 8

Signs with a late medial distribution



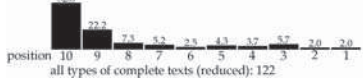
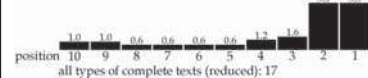
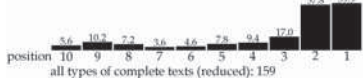
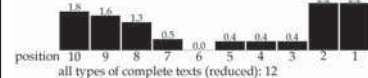
4. Doubled Signs

Signs can appear sometimes doubled, e.g. sign 90 is doubled to sign 91, and sign 820 doubled is sign 821. Double signs are not coded in the inscriptions as a pair of the single sign (like 90–90) but treated as a new sign, because it is known from other writing systems that doubling could affect the meaning. In alphabetic writing doubling a vowel often indicates a long vowel and thereby changing the meaning of the word, for example in English “red” versus “reed”, and “god” versus “good”. Not only vowels are doubled. In ancient Maya writing doubling the syllable *ku* results into the syllable *pi*, and doubling the logogram *IK* results into the syllable *ch'o*¹⁶.

The analysis of the sign position allows to check the effect of doubling in Indus writing. Sign 90 is a typical terminal sign, but sign 91 is used in any position with a maximum at initial position (Tab. 4). The second example is sign 820, a typical sign in initial position. The doubled sign 821 is still often initial but also frequently in late medial or terminal position.

Table 4

Comparison of signs and their doubled graphem

Sign	Doubled Graphem
 Sign 90  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 122	 Sign 91  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 17
 Sign 820  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 159	 Sign 821  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 12
 Sign 615  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 58	 Sign 617  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 68

¹⁶ Kettunen H., Christophe H. Introduction to Maya Hieroglyphs. URL: <http://www.wayeb.org/download/resources/wh2009english.pdf>, last access: 5.10.2010.

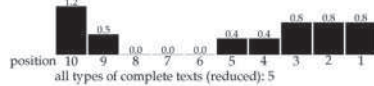
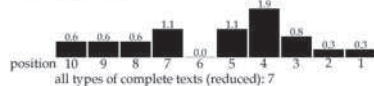
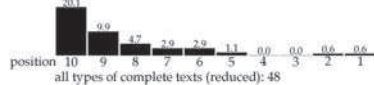
The last example is sign 615, which is mostly in late medial position but can occur anywhere except in terminal position. The doubled sign 617 is mostly in terminal position, especially in single and multiple segment texts.

The examples show, that doubling can have an effect on the sign's positional behaviour. Most likely this effect is due to a change in the grammatical function as a result of a different phonetic value. Therefore, a doubled sign should not be treated as twice the same basic sign but as a distinct sign.

5. Mirroring of signs

The comparison of sign 920 to sign 921 has already shown, that mirroring of signs can change the positional histogram (Tab. 2). Other examples are sign 407 versus 408, or 435 versus 436 (Tab. 5). In contrast, sign 526 and its mirrored sign 527 have a similar positional distribution. Other mirrored signs are too rare to be analyzed with confidence.

Table 5
Comparison of signs and their mirrored graphem

Sign	Mirrored Graphem
 Sign 407  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 56	 Sign 408  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 5
 Sign 435  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 40	 Sign 436  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 7
 Sign 526  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 11	 Sign 527  position 10 9 8 7 6 5 4 3 2 1 all types of complete texts (reduced): 48

6. Sign Functions

Another important fact to consider are different text classes. Text classes as defined by B. Wells¹⁷ distinguish between different text length (short or long) and typical sign pattern combinations¹⁸. In the following example signs 590 and 550 are analyzed for different text classes.

The histogram of positional frequency of sign 590 for all complete texts shows a maximum at position 6 (Fig. 9a). For Short Patterned texts (Fig. 9b) the maximum is at about the same position (6–7). If the frequency is counted only for Long Pattered texts (LP) a steep maximum at position 8 is shown (Fig. 9c). For Multiple segmented texts (MS) sign 590 shows a typical behaviour like an Initial sign (Fig. 9d), and in Long complex texts (LC) the positional distribution is irregular (Fig. 9e). Therefore, sign 590 has different syntactic functions for different text classes and the histogram of all texts shows the sum of the frequency curves from different text classes. This means, that complex positional frequency curves are most likely a mixture of different positional frequency curves and must be analyzed for different text classes separately.

Sign 550 also appears often in first position in about 46% of all complete texts ($n = 82$), but in all other texts the sign can occupy any sign position (Fig. 10a). Its first (absolute) maximum is at position 1, but its second (local) maximum is at position 7 to 8.

The second maximum at is highlighted in the histogram for short and partial patterned texts (Fig. 10b and c). This means, that sign 550 has most likely two different functions or values depending on its syntactic position. B. Wells¹⁹ suggest, that sign 550 is a syllabic sign, because the frequent sign pair 527–550 ($n = 28$) functions like one sign.

In two complete texts from Lothal (L–106, L–229) it is used as a singleton, suggesting to be used as a logogram. At the present, I think, we have not enough data at hand or knowledge about the syntax to decide at which position sign 550 has which function, but it has more than one function or phonetic value.

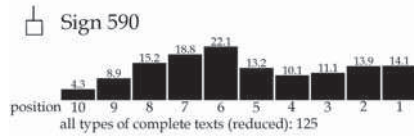
¹⁷ Wells B. *Epigraphic Approaches...* P. 134.

¹⁸ Fuls A. *Entwicklung...* Tab. 2.

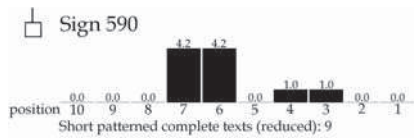
¹⁹ Wells B. *Epigraphic Approaches...* P. 152.

Fig. 9

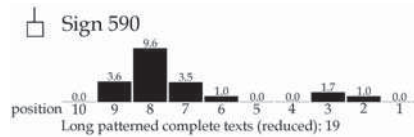
Histograms of sign 590 for different text classes



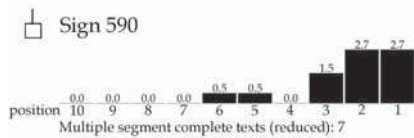
(a) all complete texts



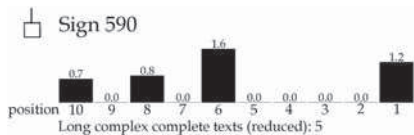
(b) short patterned texts



(c) long patterned texts



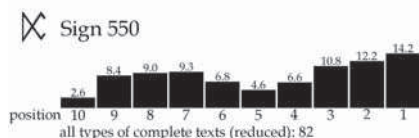
(d) multiple segment texts



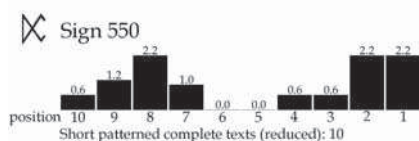
(e) long complex texts

Fig. 10

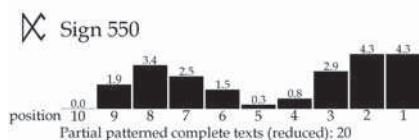
Histogram of sign 550 with two maxima at position 1 and 8 in various text classes



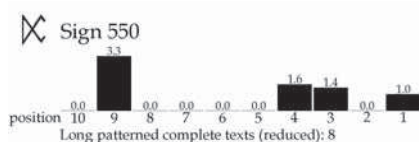
(a) all complete texts



(b) short patterned texts



(c) partial patterned texts



(d) long patterned texts

7. Summary

The normalized weighted sign position allows to summarize texts of different length to calculate a histogram of positional sign behaviour. The method distinguish different text length in that short texts have a smaller weight as longer texts. Positional preferences of signs can

now easily determined and it allows the identification of allographs by comparing positional histograms of similar graphemes.

The positional analysis results into different classifications of Indus signs. Signs can be classified as Initial, Early-Medial, Mid-Medial, Late-Medial, Terminal signs, constant or constant but not initial or not terminal. The histograms of Post-Terminal signs shows no difference to Terminal signs and can only be distinguished by detailed structural analysis.

The positional analysis shows, that doubling, mirroring, hatching, or marking of Indus signs can change their positional classification. Most likely, this indicates a different function or phonetic value of the modified signs.

Signs with a similar positional distribution can now be grouped together. In each positional group signs share the same preferred text position and thereby the same syntactic position. Further analysis need to be done to establish positional groups of signs with more confidence, since positional analysis highlights syntagmatic but not paradigmatic relationships of signs.

Positional analysis will help to distinguish between logograms and syllables. Logograms and grammatical markers should fall on restricted syntactic positions (depending on the syntax of the Indus language), while syllables should have an almost constant positional distribution. Signs 382, 790, 832, and 892 are good candidates of syllables.

Many signs have a different preferred position depending on the text class. This indicates, that they can have more than one grammatical function or phonetic value. Therefore, future analysis of the Indus script must differentiate between several text classes and deal with the possibility of polyvalent signs. Each text class has specific features like typical sign pattern of initial and terminal signs or text length²⁰ and must be analyzed separately. This is important, since previous statistical tests often analyzed all Indus texts together²¹.

²⁰ Wells B. *Epigraphic Approaches To Indus Writing*. Oakville; Oxf., 2011.

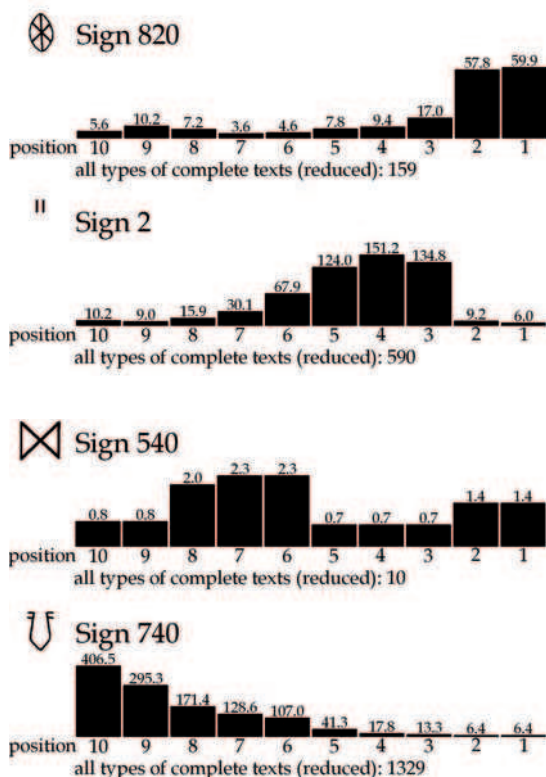
²¹ Yadav N., Vahai M. N., Mahadevan I., Joglekar H. A statistical approach...; Rao R. P. N., Yadav N., Mayank N., Vahia H., Joglekar R. A., Mahadevan I. A Markov model of the Indus script // *Proceedings of the National Academy of Sciences (PNAS)*. 2009. Vol. 106. P. 13685–13690; Sinha S., Ashraf Md. I., Raj K. P., Wells B. K. Network analysis of a corpus of undeciphered indus civilization inscriptions indicates syntactic organization // *Computer Speech and Language*. 2010. Vol. 25. P. 639–654.

The position of signs and sign sequences in a specific text class are more predictable than analyzing all text classes together, and thereby mixing different contexts or text structures.

Texts can be analyzed for the positional behaviour of its signs (Fig. 11). In patterned texts the positional histogram of each sign corresponds to its syntactic position. This allows to compare normal sign pattern against exceptions (labeled complex texts), or to search for writing errors.

Fig. 11

Positional histogram of signs in the text +740-540-002-820+ (seal M-1088). It is a typical example of a short patterned text with Initial cluster 002-820 and Terminal sign 740



Multiple line texts are excluded in the calculation of positional histograms. The signs in multiple line texts can now be analyzed to determine the reading direction and to investigate questions about the relationship of multiple lines of text to each other. Does they form a series of independent phrases or one long sentence?

Even if we are, at the present, far away from understanding the syntax of Indus writing, its grammatical rules and the underlying language, positional analysis will help to solve these questions. Each language consists of different classes of words, affixes or grammatical marker. It is time to search for them by analyzing the syntactic position of signs.

Summary

A. Fuls

Positional Analysis of Indus Signs

The Indus culture (2600 to 1700 BC) developed a writing system that remains undeciphered until today. Indus inscriptions can be found on different artefacts like seals, tablets, pots, bangles, tags, and other types. Most inscriptions are short with a mean text length of 4 signs. They are mostly read from right to left. The analysis is based on an updated sign list from B. Wells (2011) with 695 signs and an electronic text corpus of 3723 complete texts with 13556 signs.

Sign sequences show a distinct usage pattern. Some signs are typically found in initial position, while other are mostly terminal, but many signs can only be classified as medial, which does not tell us the details of these distributions.

A new method is developed to classify each sign's positional behaviour and to differentiate between different medial positions. The method calculates a Normalised Weighted Sign Position, which reduces the position of each sign to a standard text length of ten signs. To avoid an over-weighting of signs from short texts, each sign position is weighted linearly to the original text length. This allows a more detailed analysis and classification of Indus signs for all texts, as well as for specific text classes based on sign behaviour (B. Wells 2011). As a result it is possible to classify Indus signs as initial, terminal, post-terminal, constant, con-

stant but not initial, early medial, mid medial, or late medial. The method can also be used to show that often graphically similar signs differ in their preferred positional behaviour and are not stylistic variations (allographs) but distinct signs (graphemes). For example, it is necessary to distinguish between initial signs 692, 817, 820, 861, and 920 and their graphically similar counterparts 690, 790, 824, 850, and 921. Therefore, sign lists published by A. Parpola (386 signs) and I. Mahadevan (419 signs), who have merged graphically similar signs with different positional behaviour, are not useful in the analysis of Indus writing.

The application of the Normalised Weighted Sign Position method indicates, that doubling as well as mirroring a sign can have an effect on the sign's positional behaviour. Most likely this effect is due to a change in the grammatical function as a result of a different phonetic value. Therefore, a doubled sign should not be treated as twice the same basic sign but as a distinct sign. Additionally, mirrored signs should be treated as distinct signs.

B. Wells (2011) introduced the definition of different text classes. Depending on text length (short or long) and typical sign pattern combinations he distinguishes between short, long, and partial patterned, short and long complex, single segment and multiple segmented texts as well as specific texts like numerical texts and two or three line texts. The positional histograms of signs derived from different text classes indicate, that Indus signs can have more than one preferred text position depending on the content and/or the structure of the texts (polyvalence). This is important to emphasise, because very often previous statistical tests of Indus signs have analysed all texts together without considering geographic, temporal variations or different contexts or text structures. With the introduction of this new method we can begin to understand the full structure of Indus writing.

Резюме
А. Фульс

Позиционный анализ знаков индского письма

В ходе развития протоиндийской цивилизации (2600–1700 гг. до н.э.) возникла система письма, которая до настоящего времени остаётся нерасшифрованной. Надписи находят на различных артефактах: печатях, табличках, горшках, браслетах, бирках и др. Боль-

шинство надписей представляют собой короткие тексты со средней длиной в 4 знака. Чаще всего они читаются справа налево.

Анализ основан на обновлённом списке знаков из публикации Б. Уэллса (2011), включающей 695 единиц, и электронном корпусе текстов, который состоит из 3723 полных текстов с 13 556 знаками.

Последовательность знаков показывает некоторые особенности их использования. Некоторые из них встречаются преимущественно в начале текста, другие — в конце, но большинство — только в середине. Но одна только констатация этого ещё ничего не объясняет.

Нами был разработан новый метод, целью которого является классификация позиционного поведения каждого знака и определения различий между теми из них, которые находятся в середине текста. Данный метод предполагает подсчёт позиций «стандартного взвешенного знака», сократив число позиций каждого знака внутри текстов стандартной длиной в десять знаков. Чтобы избежать их «перевеса» в более кратких текстах, позиция каждого из них взвешивалась по линейному закону с учётом изначальной длины текста.

Это позволило провести более детальный анализ знаков индского письма и, основываясь на поведении знака, разработать их классификацию для всех текстов, а также для специфических видов текстов (Б. Уэллс 2011). В результате оказалось возможным классифицировать данные знаки как начальные, конечные, постконечные, постоянные, постоянные-не-начальные, ранне-средние, средне-средние и поздне-средние.

С помощью данного метода можно увидеть, насколько часто графически схожие знаки различаются по своему предпочтительному позиционному поведению и являются не стилистическими вариантами (аллографами), а отдельными знаками (графемами). Например, необходимо различать начальные знаки 692, 817, 820, 861 и 920 и их графически похожие аналоги 690, 790, 824, 850 и 921. Таким образом, списки знаков, опубликованные А. Парпола (386 знаков) и И. Махадеваном (419 знаков), которые объединяют графически схожие знаки с различным позиционным поведением, не могут быть использованы при анализе знаков индского письма.